



Networking Fundamentals Overview

A concise synthesis of three PDFs covering **inter-VLAN routing** troubleshooting, core **file-transfer and remote-access protocols**, and essential **Cisco switched-network** concepts and commands.



Content Sources

- Section 1: Details **Inter-VLAN routing** and **troubleshooting** on Layer 3 devices from a 44-page PDF document. ([source](#))
- Section 2: Introduces **FTP**, **SCP**, and **SSH** file-transfer and remote-access protocols from a 24-page PDF document. ([source](#))
- Section 3: Summarizes **Switched Networks** fundamentals and key **Cisco IOS** commands from a 17-page PDF document. ([source](#))

Section 1: Inter-VLAN Routing Mastery

This section covers comprehensive inter-VLAN routing concepts from a 44-page PDF document on switching and routing ([source](#))

Module Overview

Module Objective: Troubleshoot inter-VLAN routing on Layer 3 devices

Topic	Objective
Inter-VLAN Routing Operation	Describe the three options for configuring inter-VLAN routing
Router-on-a-Stick Inter-VLAN Routing	Configure router-on-a-stick inter-VLAN routing
Inter-VLAN Routing using Layer 3 Switches	Configure inter-VLAN routing using Layer 3 switching
Troubleshoot Inter-VLAN Routing	Diagnose common inter-VLAN configuration problems

Understanding Inter-VLAN Routing

Inter-VLAN routing is the process of forwarding traffic from one VLAN to another VLAN, enabling communication between hosts in different VLANs.

Key Principle: VLANs segment a Layer 2 switched network, and hosts in different VLANs **cannot communicate** unless a **router** or **Layer 3 switch** provides routing services.

Three Routing Options

Option	Description	Scalability
Legacy Inter-VLAN routing	Multiple physical router interfaces, each linked to a VLAN	Low – limited by router's physical ports
Router-on-a-Stick	One router physical interface operates as an 802.1Q trunk; subinterfaces represent each VLAN	Acceptable for small/medium-size networks; practical limit $\approx$ 50 VLANs
Layer 3 Switch (SVIs)	Switched Virtual Interfaces (SVIs) on a multilayer switch act as routed interfaces per VLAN	Most scalable; hardware-based switching and routing

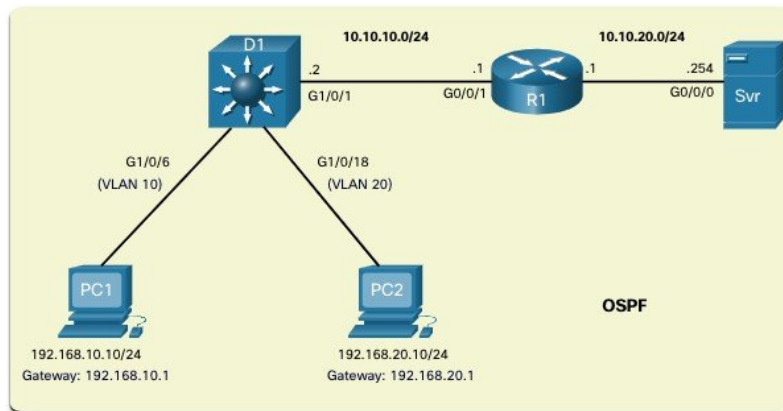
Router-on-a-Stick Configuration

Core Concept: A single router Ethernet interface configured as an **802.1Q trunk** linked to a trunk port on a Layer 2 switch.

Subinterface Configuration Example:

```
interface GigabitEthernet0/0/1.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0/1.20
  encapsulation dot1q 20
  ip address 192.168.20.1 255.255.255.0
!
```

```
interface GigabitEthernet0/0/1
no shutdown
```



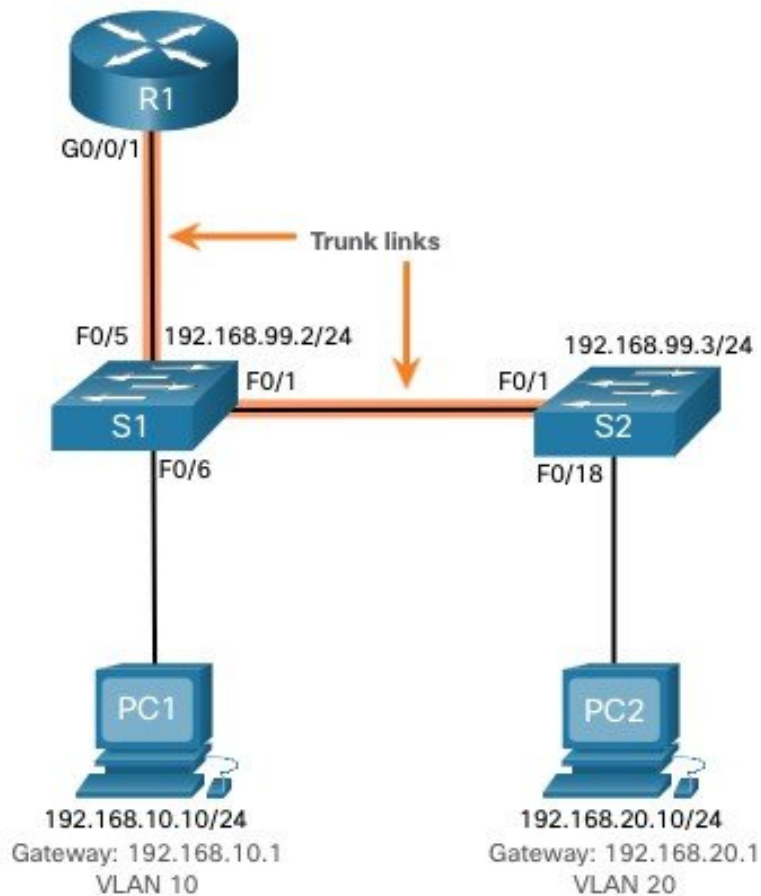
Layer 3 Switch Configuration

Advantages:

- Hardware-based routing (faster than router-on-a-stick)
- No external router link needed
- Supports EtherChannel trunk links
- Lower latency
- Campus LAN suitability

SVI Configuration Example:

```
interface vlan 10
  ip address 192.168.10.1 255.255.255.0
!
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
!
ip routing
```



Troubleshooting Common Issues

Issue Type	Typical Causes	Remediation
Missing VLANs	VLAN not created or not allowed on trunk	show vlan brief; create or re-add VLAN
Switch Trunk Port Issues	Trunk not configured or mismatched encapsulation	show interface trunk; verify configuration
Switch Access Port Issues	Wrong VLAN assignment or host in wrong subnet	show interfaces switchport; check assignment
Router Configuration Issues	Subinterface IP mis-configured or wrong VLAN	show ip interface brief; verify subinterface settings

Section 2: 📁 File Transfer & Remote Access Protocols

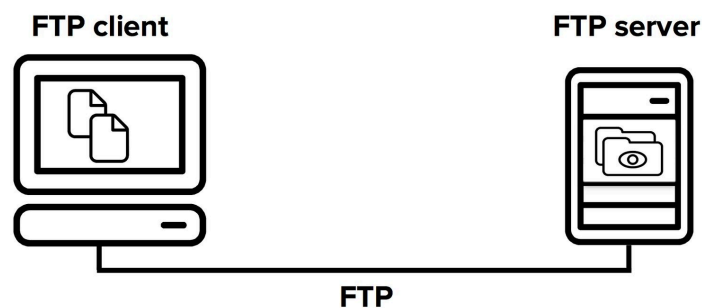
This section explores file transfer and remote access protocols from a 24-page PDF document ([source](#))

File Transfer Protocol (FTP)

FTP is the simplest way to exchange files over the Internet, supporting both **uploading** (client to server) and **downloading** (server to client).

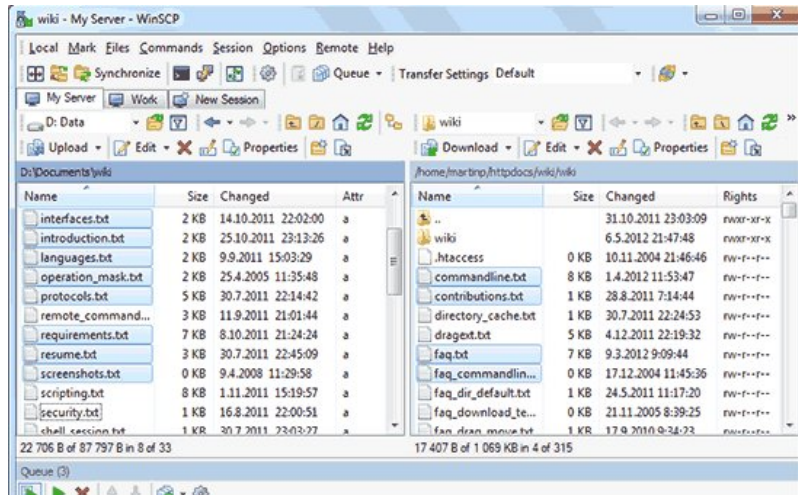
Basic Architecture:

- Requires an **FTP client** program on the user's machine
- Connects to an **FTP server** running a daemon (like VSFTPD in Linux)
- Operates in **client/server** model



Common FTP Clients

GUI Client	Platform	Notes
WinSCP	Windows	Open-source file manager
Cyberduck	Mac/Windows	Multi-protocol support
Transmit	macOS	Commercial client
FileZilla	Cross-platform	Widely used, free



FTP Protocol Details

Operation Mechanics:

1. **Control Channel** - stays open on **TCP port 21** for commands
2. **Data Channel** - opened on demand for each data transfer
3. Supports auxiliary commands: create/delete directories, list files, rename files

FTP vs. HTTP Comparison:

Aspect	FTP	HTTP
Directionality	Two-way	One-way
Typical Use	File transfer	Web resources
Encoding	Binary	MIME
Ports	21 (control)	80
Result	Files saved to disk	Files rendered in browser

Trivial File Transfer Protocol (TFTP)

Feature	Description
Purpose	Lightweight file transfer
Transport	UDP port 69
Typical Use	Router configuration backup/restore
Security	No authentication, clear text

Resource Needs	Minimal
----------------	---------

Secure File Copy (SCP)

- Command-line program for copying files between hosts
- Relies on **SSH** protocol for transport
- Inherits SSH's encryption and authentication
- Pre-installed on most Linux/Unix systems

Hypertext Transport Protocol Secure (HTTPS)

HTTPS = HTTP + SSL/TLS, using **TCP port 443**

SSL Certificate Overview:

- Identifies the organization
- Enables encryption of sensitive data
- Contains **public key** and **private key**

How Encryption Works:

1. Server presents SSL certificate (public key) to client
2. Client encrypts session data with public key
3. Only server with private key can decrypt data

Remote Access Protocols

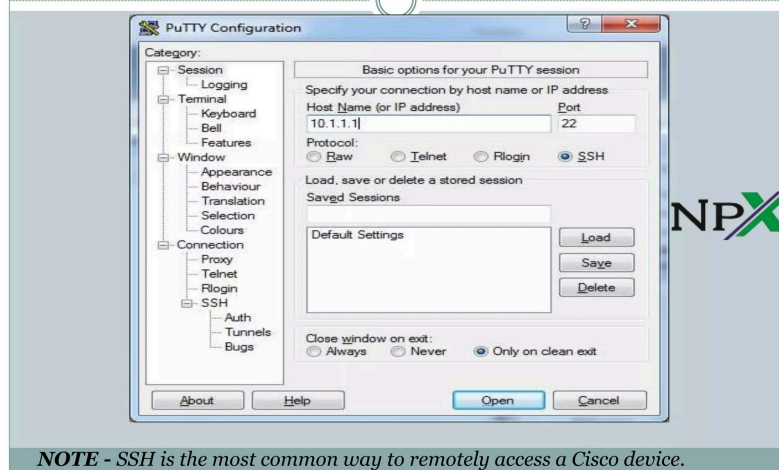
Telnet

- Virtual-terminal protocol for remote device access
- Operates over **TCP port 23**
- **Clear-text** transmission (no encryption)
- Largely replaced by SSH due to security concerns

Secure Shell (SSH)

- Provides secure, encrypted remote access
- Uses **TCP port 22**
- Relies on **public-key cryptography**
- Requires SSH client (like PuTTY) and SSH server

Here is an example of creating an SSH session using Putty, a free SSH client:



NOTE - SSH is the most common way to remotely access a Cisco device.

Protocol Summary Table

Protocol	Primary Use	Transport	Port(s)	Security
FTP	File transfer	TCP	21	Plain text
TFTP	Simple file transfer	UDP	69	No authentication
SCP	Secure file copy	SSH (TCP)	22	Encrypted
HTTPS	Secure web delivery	TCP (TLS)	443	Encrypted
Telnet	Remote access	TCP	23	Clear text
SSH	Secure remote access	TCP	22	Encrypted

Section 3: Cisco Switch Configuration Fundamentals

This section examines switched networks and switch configuration from a 17-page PDF document ([source](#))

Introduction to Switched Networks

The chapter focuses on **Cisco IOS command set** used to configure and manage switches, emphasizing how switches operate within a LAN and differ from routers.

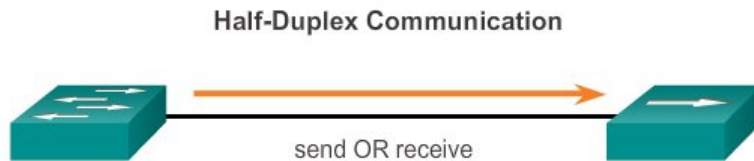
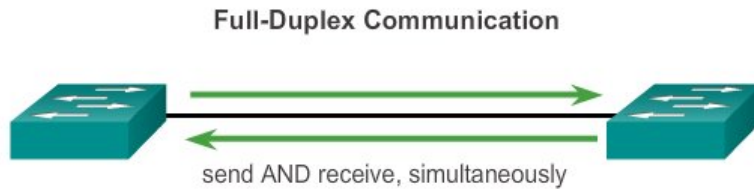
Core Cisco Switch IOS Commands

Cisco Switch IOS Commands	
Enter global configuration mode.	S1# configure terminal
Configure the default gateway for the switch.	S1(config)# ip default-gateway 172.17.99.
Return to the privileged EXEC mode.	S1(config-if)# end
Save the running config to the startup config.	S1# copy running-config startup-config

Essential Commands:

Action	IOS Command
Enter global configuration mode	configure terminal
Enter interface configuration mode	interface
Assign IP address to SVI	ip address
Enable interface	no shutdown
Return to privileged EXEC mode	end
Save configuration	write memory
Configure default gateway	ip default-gateway

Duplex Communication Fundamentals



Communication Modes:

- **Full-duplex** - device can **send and receive simultaneously**
- **Half-duplex** - device can **either send or receive, but not both**

Configuration Example:

```
interface FastEthernet0/1
  speed 100
  duplex full
```

Preparing Switch for Remote Management

Requirements for Remote Access:

- **IP address** - gives switch unique identity on LAN
- **Subnet mask** - defines IP network range
- **Default gateway** - enables management traffic to leave local subnet

SVI Configuration Steps:

1. configure terminal
2. interface vlan 1
3. ip address 192.168.1.2 255.255.255.0
4. no shutdown
5. ip default-gateway 192.168.1.1 (if needed)
6. write memory

Secure Remote Access - SSH

SSH vs. Telnet Comparison:

Property	SSH	Telnet
Encryption	Yes (strong)	No
Default port	22	23
Recommended for	All management	Legacy only

SSH Configuration:

```
crypto key generate rsa
ip domain-name example.com
line vty 0 4
  transport input ssh
  login local
username admin privilege 15 secret
```

Switch Port Security

Security Measures:

1. **Disable unused ports** - interface range FastEthernet0/1 - 24, shutdown
2. **Network Time Protocol (NTP)** - synchronizes clocks across network

NTP Configuration:

```
ntp server 192.0.2.10
ntp server 192.0.2.20
```

Verification Commands:

- show ntp status - displays synchronization state
 - show clock - displays current system clock
 - show ip ssh - displays SSH server status
 - show users - lists active sessions
-

Putting it All Together

Unified View of LAN Design & Management

- **Layer 2 switching** supplies the physical fabric; **speed/duplex** settings (speed 100, duplex full) guarantee collision-free links.
- **Layer 3 capabilities** (router-on-a-stick subinterfaces, SVIs on a multilayer switch, or routed ports) enable **inter-VLAN routing**, turning isolated VLANs into a single logical network.
- **Remote-access & file-transfer protocols** (SSH, SCP, HTTPS, FTP/TFTP) are the means by which administrators move configurations, firmware, or data across that routed fabric.

Inter-VLAN routing is the bridge that lets management traffic (SSH, SCP, etc.) travel between VLAN-segmented devices.

How Configuration Steps Interlock

1. **Prepare the switch**
 - Assign an IP to an SVI (interface vlan 1 → ip address 192.168.1.2 255.255.255.0).
 - Set a **default gateway** so the switch can be reached from other subnets.
2. **Enable Layer 3 routing** (if required)
 - Global command ip routing.
 - Create **SVIs** for each VLAN (interface vlan 10, ip address 192.168.10.1 ...).
 - Or configure a **router-on-a-stick** subinterface (interface GigabitEthernet0/0/1.20, encapsulation dot1q 20).
3. **Secure management access**
 - Generate RSA keys, set domain name, and restrict VTY lines to **SSH** only.
 - Verify with show ip ssh and show users.
4. **Deploy file-transfer services**
 - Use **SCP** (leveraging the same SSH session) for encrypted copies of configs or IOS images.
 - Reserve **TFTP** for quick, unauthenticated transfers in a trusted lab environment.
 - Prefer **HTTPS** for web-based device portals; disable plain **HTTP**.

```
crypto key generate rsa
ip domain-name corp.example
line vty 0 4
  transport input ssh
  login local
username admin privilege 15 secret StrongPwd!
```

Protocol & Routing Option Comparison

Feature	Router-on-a-Stick	Layer 3 Switch (SVI)	FTP	SCP	HTTPS
Scalability	~50 VLANs (single trunk)	Hundreds, hardware-based	N/A	N/A	N/A
Latency	Higher (router CPU)	Low (ASIC forwarding)	Depends on TCP	Encrypted, slightly higher	Encrypted, TLS overhead
Security	Relies on router ACLs	Can apply port security, ACLs	Plain-text	Encrypted (SSH)	Encrypted (TLS)
Typical Use	Small-to-mid sites	Enterprise campus core	Bulk file exchange	Secure admin/file copy	Secure web services

Security Practices That Span All Topics

- **Disable unused ports** (interface range ... shutdown) to shrink the attack surface.
- **Enforce SSH only** on VTY lines; block Telnet (transport input ssh).
- **Use NTP** (ntp server 192.0.2.10) so logs from routers, switches, and servers share a trusted timestamp.
- **Prefer encrypted transfers** (SCP, HTTPS) over clear-text protocols (FTP, TFTP, Telnet).

Checklist for a Fully Integrated LAN

1. Verify VLAN creation and trunk allowance on all switches.

2. Confirm each VLAN has an active **gateway interface** (SVI or router subinterface).
3. Test inter-VLAN connectivity (ping 192.168.20.1 from a host in VLAN 10).
4. Ensure **SSH** is reachable (ssh admin@192.168.1.2).
5. Perform a secure file copy (scp config.cfg admin@192.168.1.2:/flash).
6. Review NTP sync (show ntp status).

By weaving together **switch configuration fundamentals**, **inter-VLAN routing mechanisms**, and **secure protocol choices**, a network engineer can build a resilient, manageable, and scalable LAN environment.

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Section 1: Inter-VLAN Routing Mastery

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Module Overview

Module Objective: Troubleshoot inter-VLAN routing on Layer 3 devices

Topic	Objective
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Inter-VLAN Routing Operation	Describe the three options for configuring inter-VLAN routing
Router-on-a-Stick Inter-VLAN Routing	Configure router-on-a-stick inter-VLAN routing
Inter-VLAN Routing using Layer 3 Switches	Configure inter-VLAN routing using Layer 3 switching
Troubleshoot Inter-VLAN Routing	Diagnose common inter-VLAN configuration problems

Understanding Inter-VLAN Routing

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Key Principle: VLANs segment a Layer 2 switched network, and hosts in different VLANs **cannot communicate** unless a **router** or **Layer 3 switch** provides routing services.

Three Routing Options

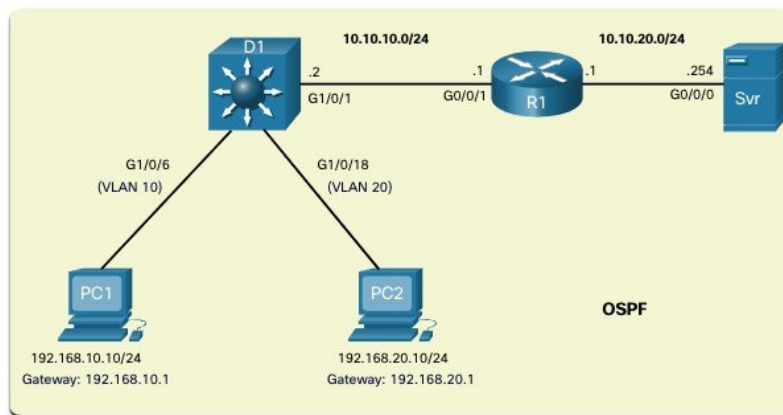
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Layer 3 Switch (SVIs)	Switched Virtual Interfaces (SVIs) on a multilayer switch act as routed interfaces per VLAN	Most scalable; hardware-based switching and routing

Router-on-a-Stick Configuration

Core Concept: A single router Ethernet interface configured as an **802.1Q trunk** linked to a trunk port on a Layer 2 switch.

Subinterface Configuration Example:

```
interface GigabitEthernet0/0/1.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0/1.20
  encapsulation dot1q 20
  ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0/1
  no shutdown
```



Layer 3 Switch Configuration

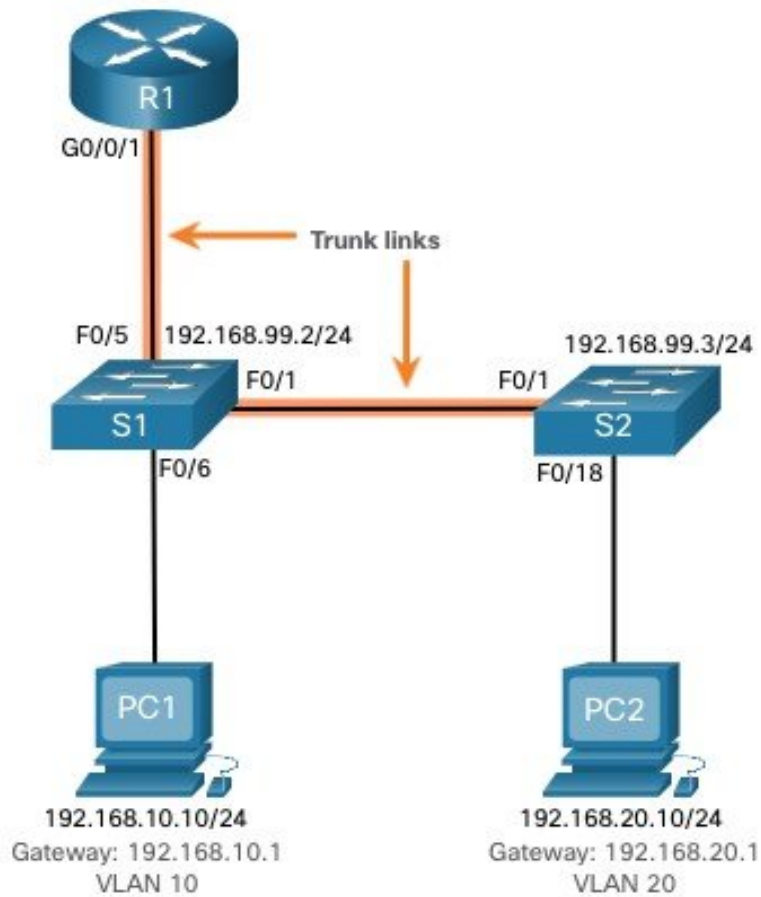
Advantages:

- Hardware-based routing (faster than router-on-a-stick)
- No external router link needed
- Supports EtherChannel trunk links
- Lower latency
- Campus LAN suitability

SVI Configuration Example:

```
interface vlan 10
  ip address 192.168.10.1 255.255.255.0
!
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
```

```
!  
ip routing
```



Troubleshooting Common Issues

Issue Type	Typical Causes	Remediation
Missing VLANs	VLAN not created or not allowed on trunk	show vlan brief; create or re-add VLAN
Switch Trunk Port Issues	Trunk not configured or mismatched encapsulation	show interface trunk; verify configuration
Switch Access Port Issues	Wrong VLAN assignment or host in wrong subnet	show interfaces switchport; check assignment
Router Configuration Issues	Subinterface IP mis-configured or wrong VLAN	show ip interface brief; verify subinterface settings

Section 2: 📁 File Transfer & Remote Access Protocols

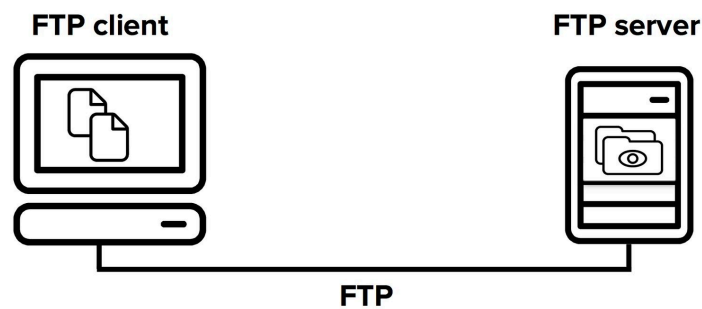
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File Transfer Protocol (FTP)

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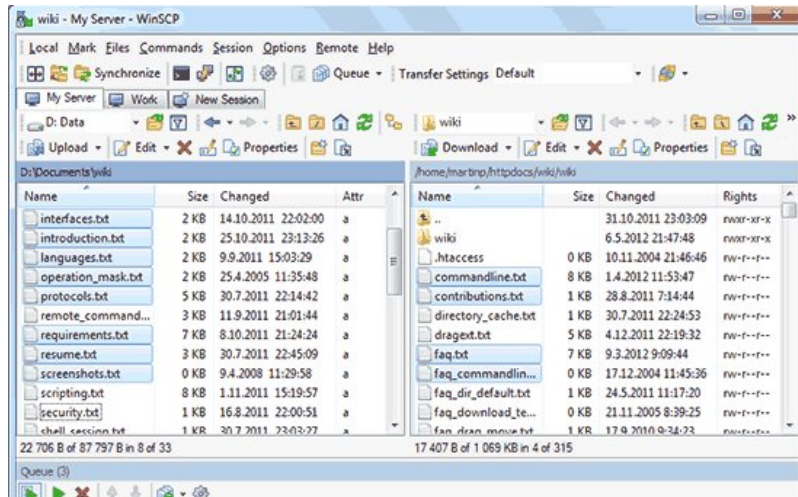
Basic Architecture:

- Requires an **FTP client** program on the user's machine
- Connects to an **FTP server** running a daemon (like VSFTPD in Linux)
- Operates in **client/server** model



Common FTP Clients

GUI Client	Platform	Notes
WinSCP	Windows	Open-source file manager
Cyberduck	Mac/Windows	Multi-protocol support
Transmit	macOS	Commercial client
FileZilla	Cross-platform	Widely used, free



FTP Protocol Details

Operation Mechanics:

1. **Control Channel** - stays open on **TCP port 21** for commands
2. **Data Channel** - opened on demand for each data transfer
3. Supports auxiliary commands: create/delete directories, list files, rename files

FTP vs. HTTP Comparison:

Aspect	FTP	HTTP
Directionality	Two-way	One-way
Typical Use	File transfer	Web resources
Encoding	Binary	MIME
Ports	21 (control)	80
Result	Files saved to disk	Files rendered in browser

Trivial File Transfer Protocol (TFTP)

Feature	Description
Purpose	Lightweight file transfer
Transport	UDP port 69
Typical Use	Router configuration backup/restore
Security	No authentication, clear text

Resource Needs	Minimal
----------------	---------

Secure File Copy (SCP)

- Command-line program for copying files between hosts
- Relies on **SSH** protocol for transport
- Inherits SSH's encryption and authentication
- Pre-installed on most Linux/Unix systems

Hypertext Transport Protocol Secure (HTTPS)

HTTPS = HTTP + SSL/TLS, using **TCP port 443**

SSL Certificate Overview:

- Identifies the organization
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- Contains **public key** and **private key**

How Encryption Works:

1. Server presents SSL certificate (public key) to client
2. Client encrypts session data with public key
3. Only server with private key can decrypt data

Remote Access Protocols

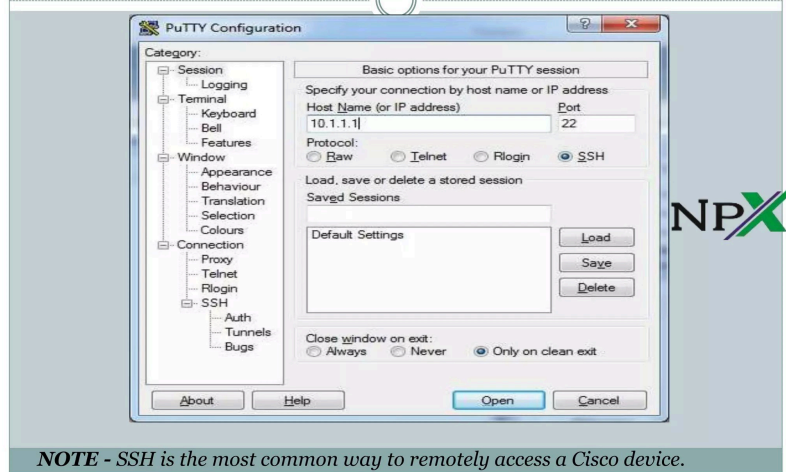
Telnet

- Virtual-terminal protocol for remote device access
- Operates over **TCP port 23**
- **Clear-text** transmission (no encryption)
- Largely replaced by SSH due to security concerns

Secure Shell (SSH)

- Provides secure, encrypted remote access
- Uses **TCP port 22**
- Relies on **public-key cryptography**
- Requires SSH client (like PuTTY) and SSH server

Here is an example of creating an SSH session using Putty, a free SSH client:



NOTE - SSH is the most common way to remotely access a Cisco device.

Protocol Summary Table

Protocol	Primary Use	Transport	Port(s)	Security
FTP	File transfer	TCP	21	Plain text
TFTP	Simple file transfer	UDP	69	No authentication
SCP	Secure file copy	SSH (TCP)	22	Encrypted
HTTPS	Secure web delivery	TCP (TLS)	443	Encrypted
Telnet	Remote access	TCP	23	Clear text
SSH	Secure remote access	TCP	22	Encrypted

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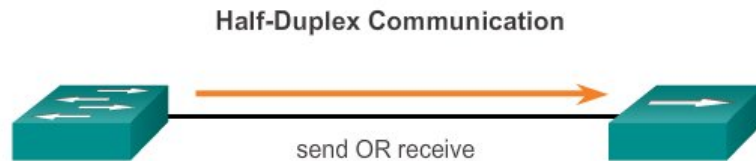
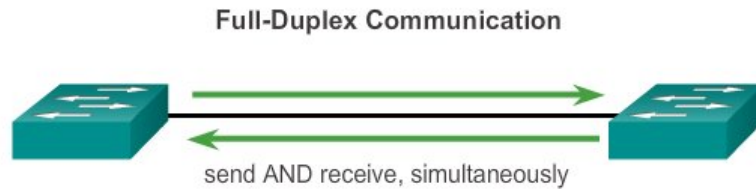
Core Cisco Switch IOS Commands

Cisco Switch IOS Commands	
Enter global configuration mode.	S1# configure terminal
Configure the default gateway for the switch.	S1(config)# ip default-gateway 172.17.99.
Return to the privileged EXEC mode.	S1(config-if)# end
Save the running config to the startup config.	S1# copy running-config startup-config

Essential Commands:

Action	IOS Command
Enter global configuration mode	configure terminal
Enter interface configuration mode	interface
Assign IP address to SVI	ip address
Enable interface	no shutdown
Return to privileged EXEC mode	end
Save configuration	write memory
Configure default gateway	ip default-gateway

Duplex Communication Fundamentals



Communication Modes:

- **Full-duplex** - device can **send and receive simultaneously**
- **Half-duplex** - device can **either send or receive, but not both**

Configuration Example:

```
interface FastEthernet0/1
  speed 100
  duplex full
```

Preparing Switch for Remote Management

Requirements for Remote Access:

- **IP address** - gives switch unique identity on LAN
- **Subnet mask** - defines IP network range
- **Default gateway** - enables management traffic to leave local subnet

SVI Configuration Steps:

1. configure terminal
2. interface vlan 1
3. ip address 192.168.1.2 255.255.255.0
4. no shutdown
5. ip default-gateway 192.168.1.1 (if needed)
6. write memory

Secure Remote Access - SSH

SSH vs. Telnet Comparison:

Property	SSH	Telnet
Encryption	Yes (strong)	No
Default port	22	23
Recommended for	All management	Legacy only

SSH Configuration:

```
crypto key generate rsa
ip domain-name example.com
line vty 0 4
  transport input ssh
  login local
username admin privilege 15 secret
```

Switch Port Security

Security Measures:

1. **Disable unused ports** - interface range FastEthernet0/1 - 24, shutdown
2. **Network Time Protocol (NTP)** - synchronizes clocks across network

NTP Configuration:

```
ntp server 192.0.2.10
ntp server 192.0.2.20
```

Verification Commands:

- show ntp status - displays synchronization state
 - show clock - displays current system clock
 - show ip ssh - displays SSH server status
 - show users - lists active sessions
-

Putting it All Together

Unified View of LAN Design & Management

- **Layer 2 switching** supplies the physical fabric; **speed/duplex** settings (speed 100, duplex full) guarantee collision-free links.
- **Layer 3 capabilities** (router-on-a-stick subinterfaces, SVIs on a multilayer switch, or routed ports) enable **inter-VLAN routing**, turning isolated VLANs into a single logical network.
- **Remote-access & file-transfer protocols** (SSH, SCP, HTTPS, FTP/TFTP) are the means by which administrators move configurations, firmware, or data across that routed fabric.

Inter-VLAN routing is the bridge that lets management traffic (SSH, SCP, etc.) travel between VLAN-segmented devices.

How Configuration Steps Interlock

1. **Prepare the switch**
 - Assign an IP to an SVI (interface vlan 1 → ip address 192.168.1.2 255.255.255.0).
 - Set a **default gateway** so the switch can be reached from other subnets.
2. **Enable Layer 3 routing** (if required)
 - Global command ip routing.
 - Create **SVIs** for each VLAN (interface vlan 10, ip address 192.168.10.1 ...).
 - Or configure a **router-on-a-stick** subinterface (interface GigabitEthernet0/0/1.20, encapsulation dot1q 20).
3. **Secure management access**
 - Generate RSA keys, set domain name, and restrict VTY lines to **SSH** only.
 - Verify with show ip ssh and show users.
4. **Deploy file-transfer services**
 - Use **SCP** (leveraging the same SSH session) for encrypted copies of configs or IOS images.
 - Reserve **TFTP** for quick, unauthenticated transfers in a trusted lab environment.
 - Prefer **HTTPS** for web-based device portals; disable plain **HTTP**.

```
crypto key generate rsa
ip domain-name corp.example
line vty 0 4
  transport input ssh
  login local
username admin privilege 15 secret StrongPwd!
```

Protocol & Routing Option Comparison

Feature	Router-on-a-Stick	Layer 3 Switch (SVI)	FTP	SCP	HTTPS
Scalability	~50 VLANs (single trunk)	Hundreds, hardware-based	N/A	N/A	N/A
Latency	Higher (router CPU)	Low (ASIC forwarding)	Depends on TCP	Encrypted, slightly higher	Encrypted, TLS overhead
Security	Relies on router ACLs	Can apply port security, ACLs	Plain-text	Encrypted (SSH)	Encrypted (TLS)
Typical Use	Small-to-mid sites	Enterprise campus core	Bulk file exchange	Secure admin/file copy	Secure web services

Security Practices That Span All Topics

- **Disable unused ports** (interface range ... shutdown) to shrink the attack surface.
- **Enforce SSH only** on VTY lines; block Telnet (transport input ssh).
- **Use NTP** (ntp server 192.0.2.10) so logs from routers, switches, and servers share a trusted timestamp.
- **Prefer encrypted transfers** (SCP, HTTPS) over clear-text protocols (FTP, TFTP, Telnet).

Checklist for a Fully Integrated LAN

1. Verify VLAN creation and trunk allowance on all switches.

2. Confirm each VLAN has an active **gateway interface** (SVI or router subinterface).
3. Test inter-VLAN connectivity (ping 192.168.20.1 from a host in VLAN 10).
4. Ensure **SSH** is reachable (ssh admin@192.168.1.2).
5. Perform a secure file copy (scp config.cfg admin@192.168.1.2:/flash).
6. Review NTP sync (show ntp status).

By weaving together **switch configuration fundamentals**, **inter-VLAN routing mechanisms**, and **secure protocol choices**, a network engineer can build a resilient, manageable, and scalable LAN environment.